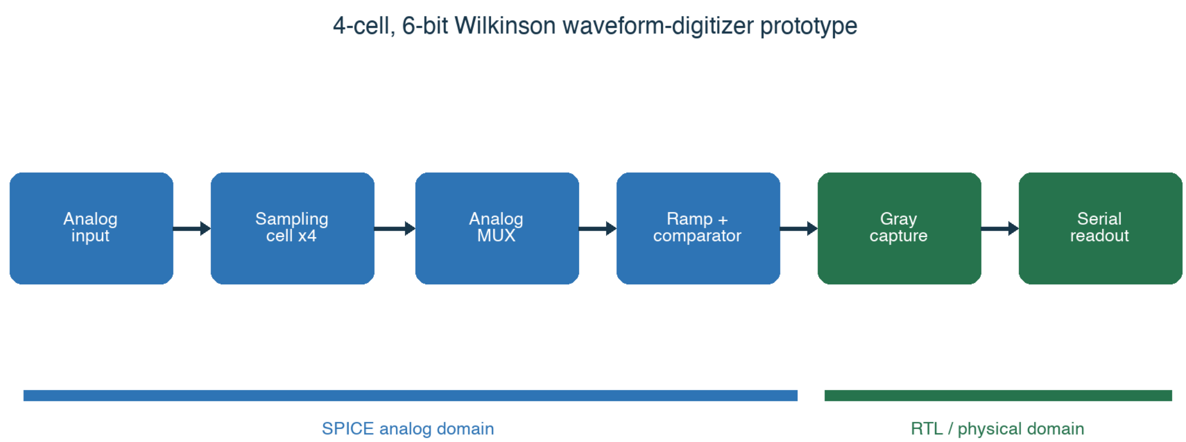
COMPLETE DESIGN FLOW

IRSX-like ASIC Prototype

GF180 Wilkinson ADC RTL-to-GDS



1

Version 1.0 | 2026-08-11

Repository: github.com/sykeisuke/asic_rd

ASIC SPICE RTL GDSII GitHub

4

- GF180 PDK MOS
- Wilkinson ADC
- SPICE RTL Gray code CDC
- RTL-to-GDS DRC/LVS/STA
- prototype tape-out-ready design

Lab	
0–2	PDK MOS sampling
3–5	comparator ramp 1/4-cell Wilkinson
6–8	controller co-sim CDC serial
9	netlist GDSII
10	blocker

0.

ASIC software process
3 log

4

		project
/ schematic		nmos_dc.sch, sampling_cell.sch
netlist	simulator	*.spice, Xschem netlist
RTL	clock digital	*.v, synthesizable Verilog
layout	silicon layer	DEF, GDSII

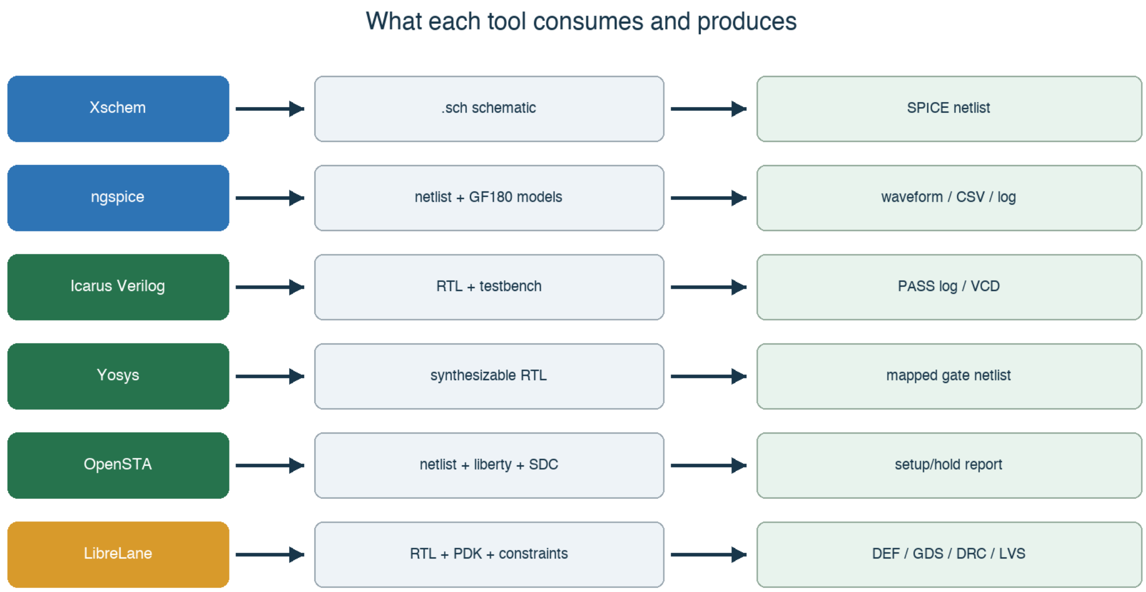
model

- SPICE model transistor nominal simulation silicon
- RTL simulation logic clock transistor-level analog
- GDS package bond wire PCB system

m	10^-3	1 mV = 0.001 V
μ / u	10^-6	1 μs = 0.000001 s
n	10^-9	1 ns
p	10^-12	1 pF, 1 ps

Lab commit hash command PASS/FAIL artifact

1.



3 tool Makefile command

Tool		
Docker Desktop	Linux Mac	container disk
Make	command target	make nmos-dc
Xschem	analog SPICE netlist	symbol wire property netlist
ngspice	transistor-level simulation	log CSV
AWK	log/CSV	script PASS
Icarus Verilog	RTL simulation	iverilog compile vvp VCD
Yosys	RTL synthesis	mapped netlist cell
OpenSTA	static timing analysis	SDC constraint setup/hold report
LibreLane	RTL-to-GDS orchestration	config.yaml metrics final GDS
KLayout/Magic/Netgen	layout DRC/LVS analog layout	layer pin DRC marker LVS mismatch

KLayout/Magic/Netgen analog layout digital physical flow LibreLane backend tool

2. Makefile

make simulator Makefile target shell script make nmos-dc scripts/run-nmos-dc.sh script Docker Xschem ngspice

```
make nmos-dc
 -> scripts/run-nmos-dc.sh
 -> docker run -v <repo>:/foss/designs
 -> xschem -n ... nmos_dc.sch
 -> ngspice -b ...spice
 -> work/nmos_id_vds.csv
```

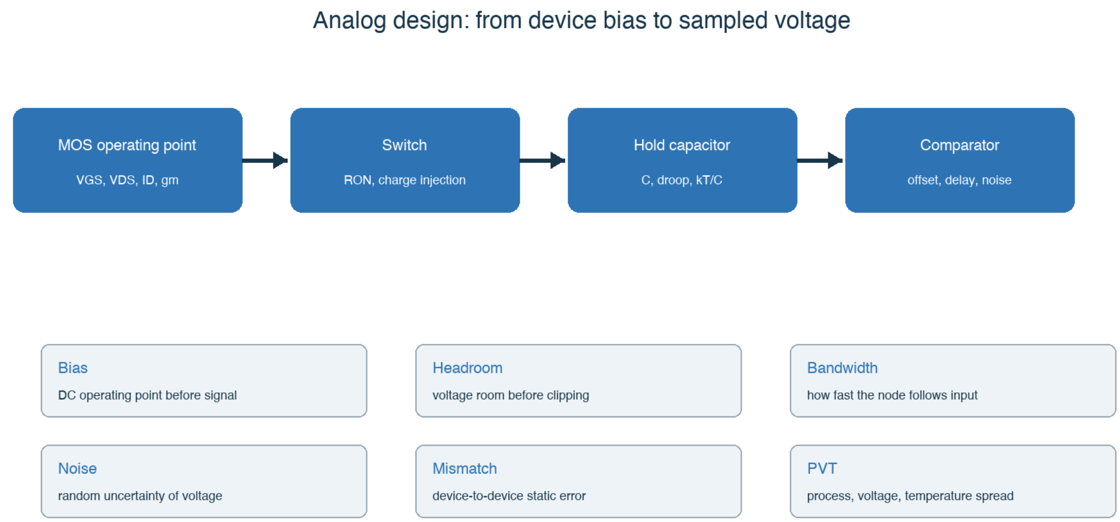
script

1. set -eu error
2. eda-common.sh EDA_IMAGE PROJECT_ROOT Docker CLI
3. docker run -v Mac repo container /foss/designs
4. container cd
5. tool option -b batch -o log
6. test/grep PASS

Target	tool	artifact
make nmos-dc	Xschem ⟶ ngspice	netlist, raw, CSV
make four-cell-cosim	ngspice ⟶ grep/AWK ⟶ iverilog/vvp	analog_stimulus.vh, cosimulation.log
make digital-top	iverilog/vvp ⟶ Yosys ⟶ OpenSTA	VCD, mapped.v, timing log
make digital-physical	LibreLane/OpenROAD ecosystem	GDS, DEF, metrics, reports

make target scripts/run-*.sh command command

3.



4 device bias sampling

digital 0/1 analog node DC operating point signal large transient noise/mismatch

3.1 MOS transistor

VGS	gate-source channel	DC sweep
VDS	drain-source headroom	I-V curve
ID	drain bias/power/	operating point
gm	gate	op result
ro	saturation	Id-Vds
W/L	device geometry mismatch	schematic property

> **body** MOS 4 device bulk/body body effect parasitic diode

3.2 Bias headroom gain bandwidth

Bias signal transistor node Headroom signal device

- DC operating point: node branch
- Gain:
- Bandwidth: gain
- Slew rate: large signal
- Settling:

RC sampling

switch ON RON hold capacitance C τ = RON·C exp(−t/τ) sampling pulse hold node settle C kT/C noise droop acquisition

C

kT/C noise droop charge sharing acquisition time switch

RON = 2 kΩ C = 1 pF τ = 2 ns 0.1% settle 7τ 14 ns first-order signal-dependent RON SPICE

3.3 noise PVT Monte Carlo

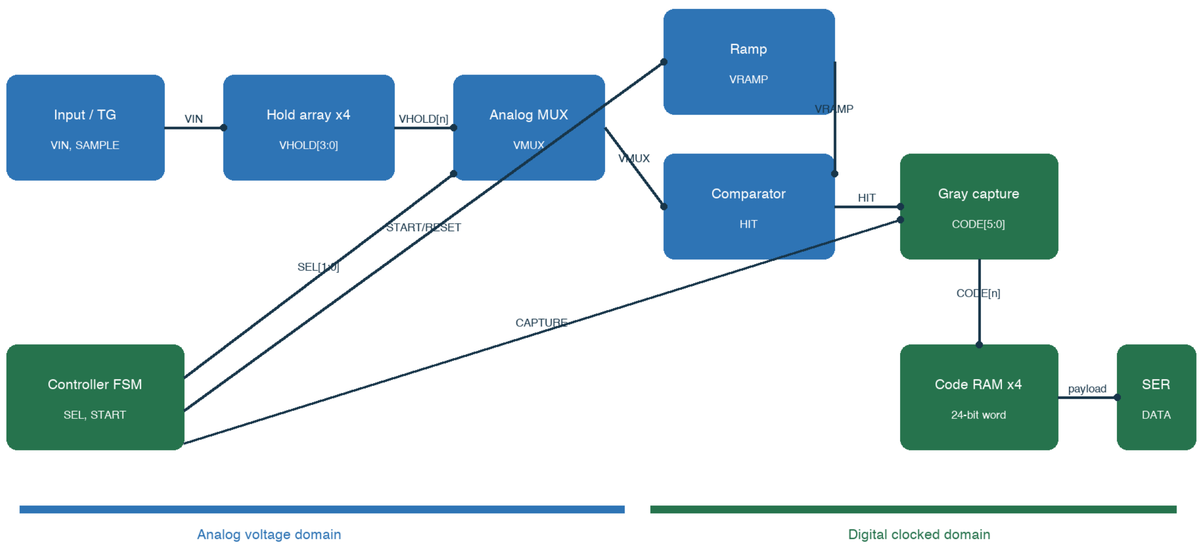
Systematic error	comparator offset ramp slope error	corner/parameter sweep calibration
Random noise	thermal noise kT/C	noise/transient noise
Mismatch	MOS	Monte Carlo mismatch
PVT	process supply temperature	corner matrix
Parasitic	layout R/C coupling	PEX simulation

Nominal simulation typical model Tape-out fast/slow device corner min/max mismatch seed extracted parasitic verification plan

7. ground
8.
9. node rail
10.
11.
12. cursor transient

4.

Prototype module connectivity and signal ownership



5 module controller analog crossing Gray capture code

Signal	Producer	Consumer	
VIN	signal source/pad	sampling transmission gate	analog voltage
SAMPLE[3:0]	controller/clock generation	sampling cells	digital control
VHOLD[3:0]	hold capacitors	analog MUX	stored analog voltage
SEL[1:0]	controller	analog MUX	selected cell index
VMUX	analog MUX	comparator	selected held voltage
VRAMP	ramp generator	comparator	time reference voltage
HIT	comparator	Gray capture/controller	asynchronous crossing event
CODE[5:0]	Gray capture	code registers	one cell result
DATA	serializer	FPGA	24-bit payload, serial

4.1 1

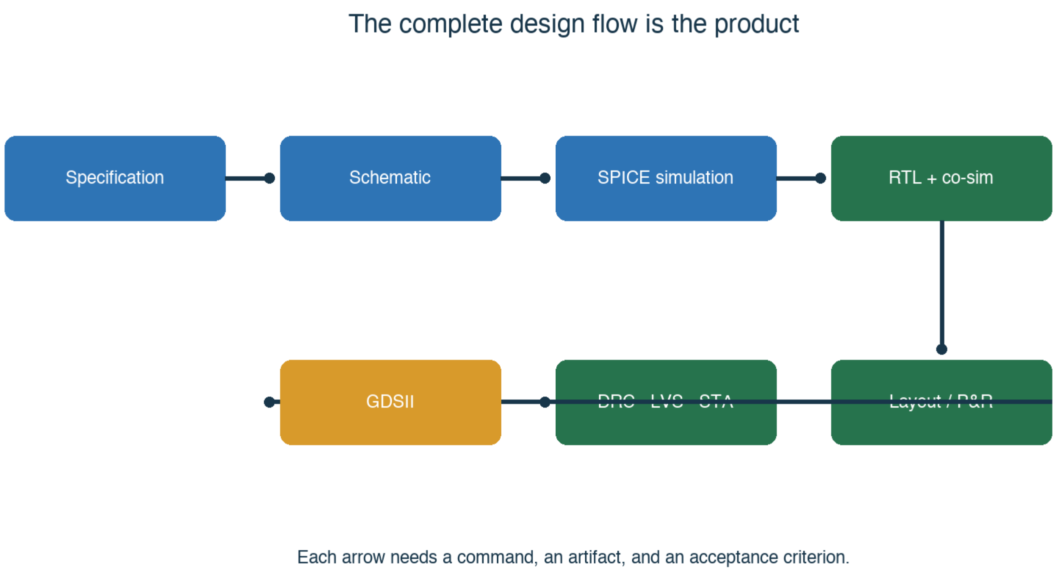
13. Track: SAMPLE[n]=1 switch ON VHOLD[n] VIN
14. Hold: SAMPLE[n]=0 switch OFF capacitor
15. Select: SEL=n VHOLD[n] VMUX
16. Settle: MUX transient
17. Reset: VRAMP counter
18. Convert: ramp Gray counter
19. Compare: VRAMP VMUX HIT edge
20. Capture: HIT Gray code binary code
21. Repeat: n=0..3
22. Readout: 4×6 bit 24-bit word serial

analog/digital

Comparator ⟶ capture	±1 code metastability	edge polarity synchronization capture convention
Controller ⟶ analog switch	short charge sharing settling	non-overlap pulse width SEL timing
Code RAM ⟶ serializer	cell bit	packing order MSB/LSB first frame valid

arrow producer consumer voltage domain clock domain active polarity reset state

5.



2 GDSII complete flow

	→	prototype
Sampling cell	VIN + SAMPLE → VHOLD	4
Analog MUX	VHOLD[3:0] + SEL → VMUX	ADC
Ramp	START → VRAMP	-
Comparator	VMUX, VRAMP → HIT	
Gray counter/capture	CLK, HIT → CODE[5:0]	bit
Serializer	CODE x4 → DATA	pad FPGA

4 storage cells 6-bit ramp/comparator 24-bit serial readout flow

Lab 0

```
git clone https://github.com/sykeisuke/asic_rd.git
cd asic_rd
git status
make check
```

5

README.md	
docs/ PROTOTYPE_SPECIFICATION.md	success criteria
docs/TOP_LEVEL_INTERFACE.md	/ I/O
docs/VERIFICATION_MATRIX.md	test
docs/TAPEOUT_BLOCKERS.md	prototype

- simulations/: Xschem/ngspice testbench
- digital/: synthesizable RTL testbench physical design
- docs/:
- work/ runs/: source of truth

- HEAD commit
- VERIFICATION_MATRIX 1 test artifact

6. Mac ✦ Docker

Mac Linux EDA GF180 PDK Docker Xschem GUI VNC macOS

23. Docker Desktop Engine Running
24. make vnc
25. http://localhost:8080/?password=abc123
26. File Manager /foss/designs/simulations/gf180_nmos_dc/nmos_dc.sch

```
cd /Users/ykeisuke/Desktop/mywork/asic_rd
make vnc
# Browser: http://localhost:8080/?password=abc123
```

image Docker Hub private image rate limit image digest tag

localhost:8080	docker ps make vnc port
Xschem PDK symbol	PDK_ROOT mount
	host path /foss/designs
com.docker.vmnetd	Docker helper macOS Gatekeeper

Gatekeeper Docker helper cleanup

6.1 Docker VNC /foss

VNC Docker Desktop Docker Desktop IIC-OSIC-TOOLS Linux container desktop Browser localhost:8080 container web/VNC server Linux desktop

```
Mac
 -> Docker Desktop
    -> container: asic-rd-gf180-vnc
       -> Linux desktop + EDA tools + PDK
       -> web/VNC server : port 80
 -> Browser localhost:8080
```

IIC-OSIC-TOOLS	Linux user environment Xschem ngspice KLayout Magic Netgen Icarus Verilog Yosys OpenSTA LibreLane/OpenROAD PDK VNC/web server
application	tool PDK version Linux image script LibreLane tool
/foss	/foss/pdks image /foss/designs clone repository Mac mount
	/foss/designs Mac repository container file container
Connected to <id>	container desktop session browser VNC
	Git commit image tag/digest Make target

volume mount

eda-vnc.sh -v "$PROJECT_ROOT:/foss/designs:rw" PROJECT_ROOT make vnc asic_rd repository

```
Student Mac: /Users/.../asic_rd
             | Docker bind mount
Container:   /foss/designs
```

git clone asic_rd directory cd make vnc /foss/designs clone mount container

7. Xschem

Xschem editor simulation engine symbol wire SPICE netlist netlist ngspice project GUI script headless netlisting

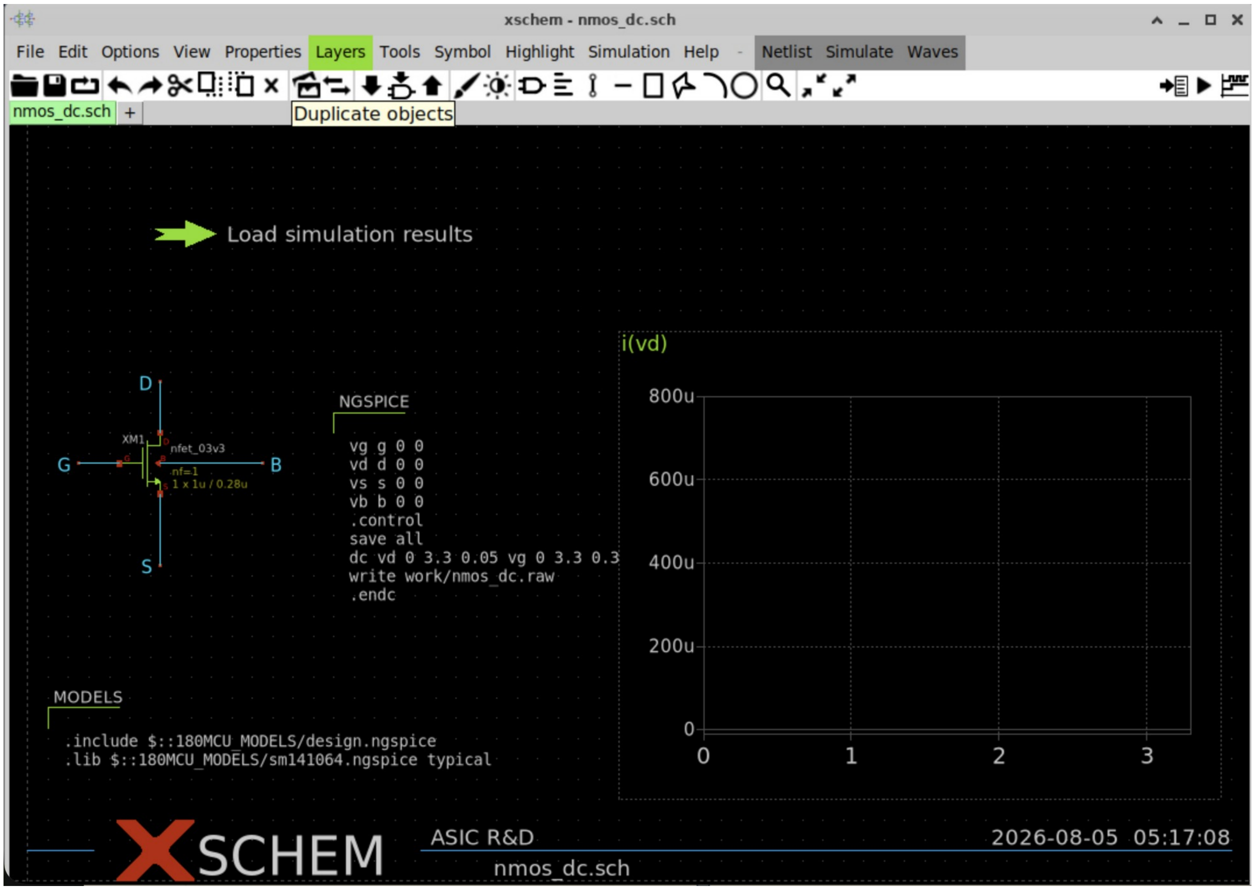
27. make vnc browser desktop
28. File Manager /foss/designs/simulations/gf180_nmos_dc/
29. nmos_dc.sch double-click Xschem File > Open
30. NMOS symbol voltage source ground simulation command
31. symbol property model W L multiplicity

- junction dot
- node label net
- ground/reference node SPICE
- MOS D/G/S/B symbol property/netlist

- testbench copy Git branch
- W L bias stimulus
-
- git diff

> Xschem shortcut version keymap menu status/help shortcut

7.1 nmos_dc.sch



6 nmos_dc.sch graph RAW

GF180 3.3 V NMOS G/D/S/B 4 W=1 µm L=0.28 µm

ngspice command VDS 0–3.3 V VGS 0–3.3 V sweep

GF180 model include typical library

i(vd) −1 * graph drain current VDS

work/nmos_dc.raw Xschem launcher

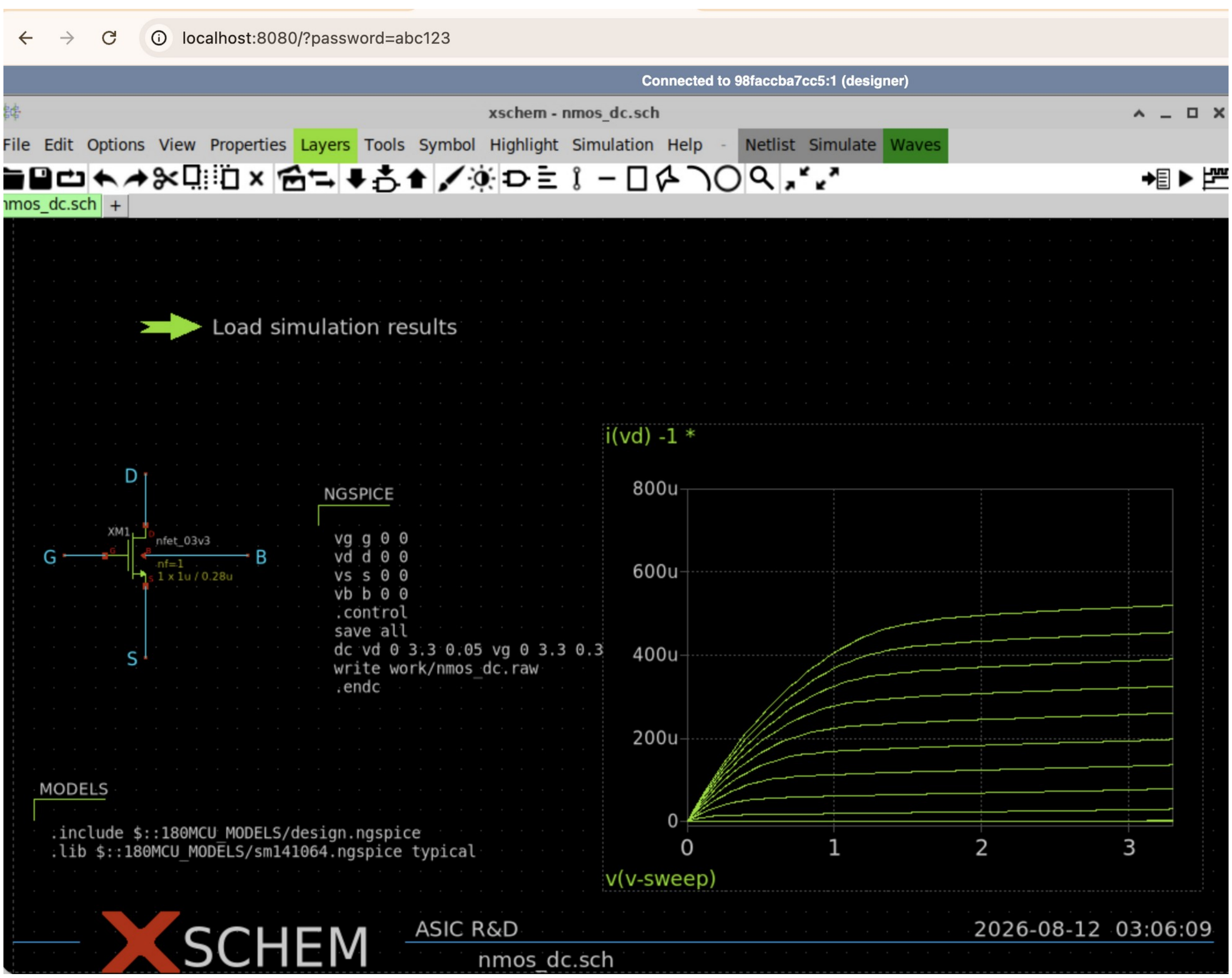
7.2

32. Mac terminal repository make nmos-dc work/nmos_dc.raw
33. Xschem Load simulation results Ctrl + launcher property
34. graph Id-Vds curve
35. curve Xschem directory /foss/designs/simulations/gf180_nmos_dc
36. Terminal ls -lh work/nmos_dc.raw file 0 byte
37. make nmos-dc Xschem Ctrl +

```
# Mac terminal
cd /Users/ykeisuke/Desktop/mywork/asic_rd
make nmos-dc

# Container terminal
cd /foss/designs/simulations/gf180_nmos_dc
ls -lh work/nmos_dc.raw
```

R W raw_read(): failed to open ... Xschem /foss/designs/simulations/gf180_nmos_dc project



7 Id–Vds VGS curve 520 μA

- VGS curve drain current
- VDS VDS VDS
- u μA 800u 800 μA
- SPICE source NMOS drain current i(vd) −1 * Xschem graph RPN
- graph curve VGS sweep
- typical model nominal DC sweep PVT

> RAW curve VGS/VDS/ID/W/L/model corner Lab 1

7.3 Netlist ngspice

make nmos-dc Xschem nmos_dc.sch work/nmos_dc_xschem.spice ngspice batch mode netlist raw data CSV

```
xschem -n -q -x --rcfile <gf180-xschemrc> \
+  -o work -N nmos_dc_xschem.spice nmos_dc.sch
ngspice -b work/nmos_dc_xschem.spice
```

Option	
-n	user config
-q	quiet startup
-x	GUI headless
--rcfile	GF180 symbol/model path
-o work	directory
-N file	netlist file
ngspice -b	batch simulation

38. terminal code 0
39. work/nmos_dc_xschem.spice device instance model include
40. work/nmos_dc.raw
41. work/nmos_id_vds.csv header
42. directory README.md RESULTS.md expected result

generated netlist netlist model node parameter

8. Digital tool

	Command	
Compile	iverilog -g2012 ...	syntax module unsupported construct
Run	vvp work/tb	assertion/PASS event
Waveform	VCD GTKWave	clock reset state capture serial data
Synthesis	yosys ... synth/abc	logic standard cell cell
Timing	sta -exit sta.tcl	clock constraint setup/hold slack
Physical	librelane ... config.yaml	floorplan routing DRC/LVS GDS

VCD

43. clock reset
44. controller state cell select hit captured code
45. cursor capture edge edge
46. unknown X impedance Z glitch
47. serializer clock/data/valid timebase

Synthesis

testbench #delay display simulation real model silicon Yosys source list run-digital-top.sh mapped netlist GF180 standard-cell instance

Lab 1 GF180 MOS

NMOS I–V bias

```
make nmos-dc
```

48. nmos_dc.sch device W/L body supply
49. testbench VGS/VDS sweep
50. make nmos-dc CSV/plot
51. VGS VDS linear saturation

- ☐ ngspice error
- ☐ simulations/gf180_nmos_dc/work/nmos_id_vds.csv
- ☐ sweep

process corner device option MPW provider

W 2 variant bias Id simulation

Lab 2 Sampling cell

track/hold acquisition droop charge injection

```
make sampling-cell
make four-cell
make four-cell-mux
```

52. SAMPLE high VHOLD VIN
53. SAMPLE edge step charge injection/feedthrough
54. hold droop rate [V/s]
55. 4-cell test cell
56. MUX cell

Cell	[V]	[V]	[mV]
0	0.490799	0.500	9.201
1	0.792128	0.800	7.872
2	1.094365	1.100	5.635
3	1.399222	1.400	0.778

prototype testbench nominal PVT/Monte Carlo

- cell sampling pulse node naming
- MUX bus SEL decode transmission gate complementary control
- capacitance leakage path simulation timestep

Lab 3 Comparator ramp

ADC

```
make comparator
make comparator-range
make comparator-offset
make ramp-generator
```

Wilkinson ramp counter capture t_cross = (V_hold − V_start) / slope ramp slope offset delay noise code

Test	
comparator	logic swing delay
range	common-mode/input range analog
offset	input-referred offset code bias
ramp	slope linearity reset LSB conversion time

6-bit

VFS LSB VFS/64 1.6 V 25 mV/LSB prototype code ordering monotonicity INL/DNL

ramp slope ±10% crossing time code

Lab 4 1-cell Wilkinson slice

sampling ramp comparator counter 1

```
make wilkinson-slice
make transfer
```

57. sampling phase input hold capacitor
58. conversion start ramp reset counter
59. VRAMP VHOLD comparator edge
60. edge counter capture end-of-conversion
61. input sweep code transfer

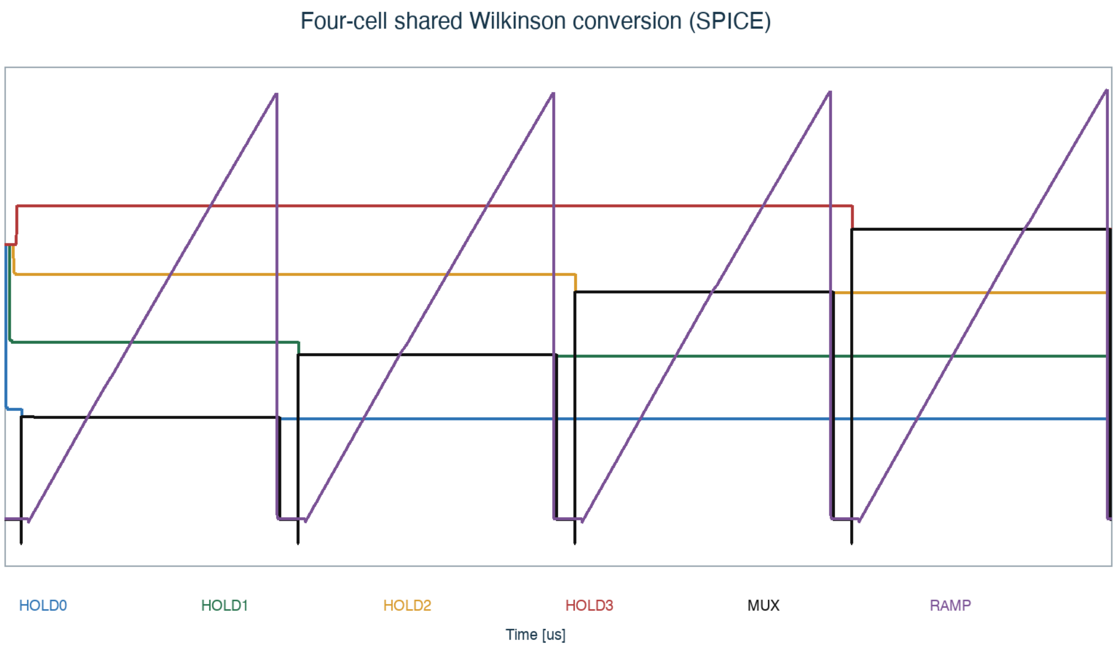
- ☐ code
- ☐ input code
- ☐ 0–63
- ☐ dead region

transient capture 1 timestep max timestep code

Lab 5 4-cell shared conversion

4 1 ramp/comparator 6-bit code

```
make four-cell-wilkinson
```



3 4-cell shared Wilkinson conversion MUX bus ramp comparator

Cell	MUX [V]	[µs]	6-bit code
0	0.456612	0.831984	16
1	0.734986	1.040260	20
2	1.014740	1.399500	27
3	1.296100	1.771850	35

code 16 < 20 < 27 < 35 sample ⟶ select ⟶ compare ⟶ capture

MUX settling cell crossing

Lab 6 Controller RTL

4-cell sequencing RTL

```
make counter
make gray-counter
make controller
make digital-top
```

62. IDLE: start valid flag clear
63. RESET_RAMP: ramp counter
64. SELECT: cell MUX settling
65. CONVERT: Gray counter comparator hit
66. CAPTURE: code register
67. NEXT/DONE: 4 cell serializer

RTL test

- reset X
- cell select one-hot binary
- capture cell
- timeout state machine
- done data_valid pulse/level testbench

simulation delay initial real synthesizable core analog testbench

Lab 7 SPICE-to-RTL

testbench end-to-end code

```
make four-cell-cosim
```

SPICE crossing event RTL testbench simulator RTL simulator AMS

contract

time unit	ns ps
edge polarity	rising/falling hit
sampling convention	edge / counter capture
code format	binary/Gray bit order width
invalid case	timeout saturation

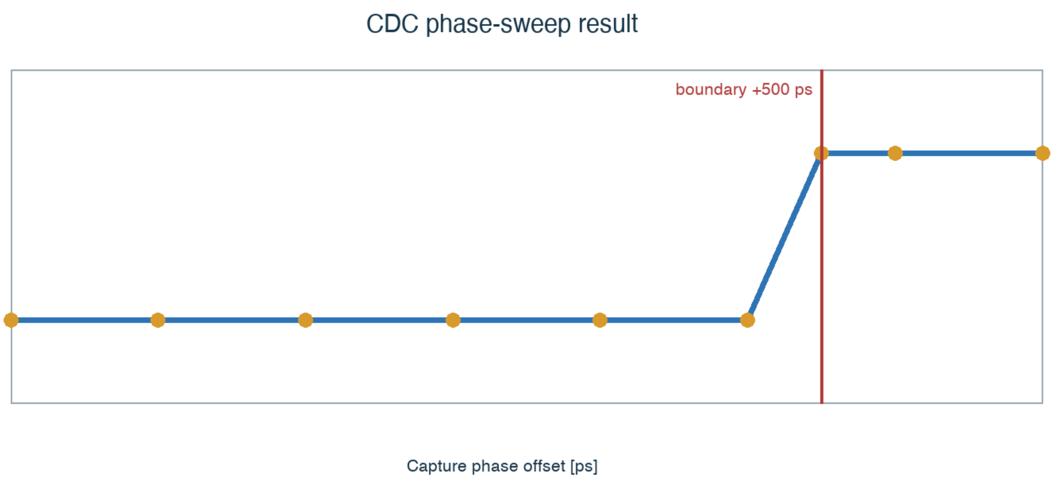
debug

68. SPICE CSV
69. event
70. testbench pulse
71. capture register enable
72. expected code ±1 sequence

Lab 8 Gray capture CDC

comparator edge counter clock

```
make phase-sweep
```



4 cell 2 phase sweep +500 ps code 27 28

binary counter 011 111⟶100 000 bit edge capture code Gray counter code 1 bit code

- +400 ps 27 +600 ps 28 ±1 code
- +500 ps cell 2 boundary setup/hold
- silicon metastability clock skew comparator delay

docs/decisions/0003-gray-comparator-capture.md

9. 24-bit serial readout

4 6-bit code 24-bit word FPGA serial pad bit order frame clock domain idle level

```
make digital-top
```

Field	
Payload	code0, code1, code2, code3
Bit order	MSB-first LSB-first
Handshake	valid/ready busy done
Clock	ASIC FPGA
Reset	reset line

logic analyzer

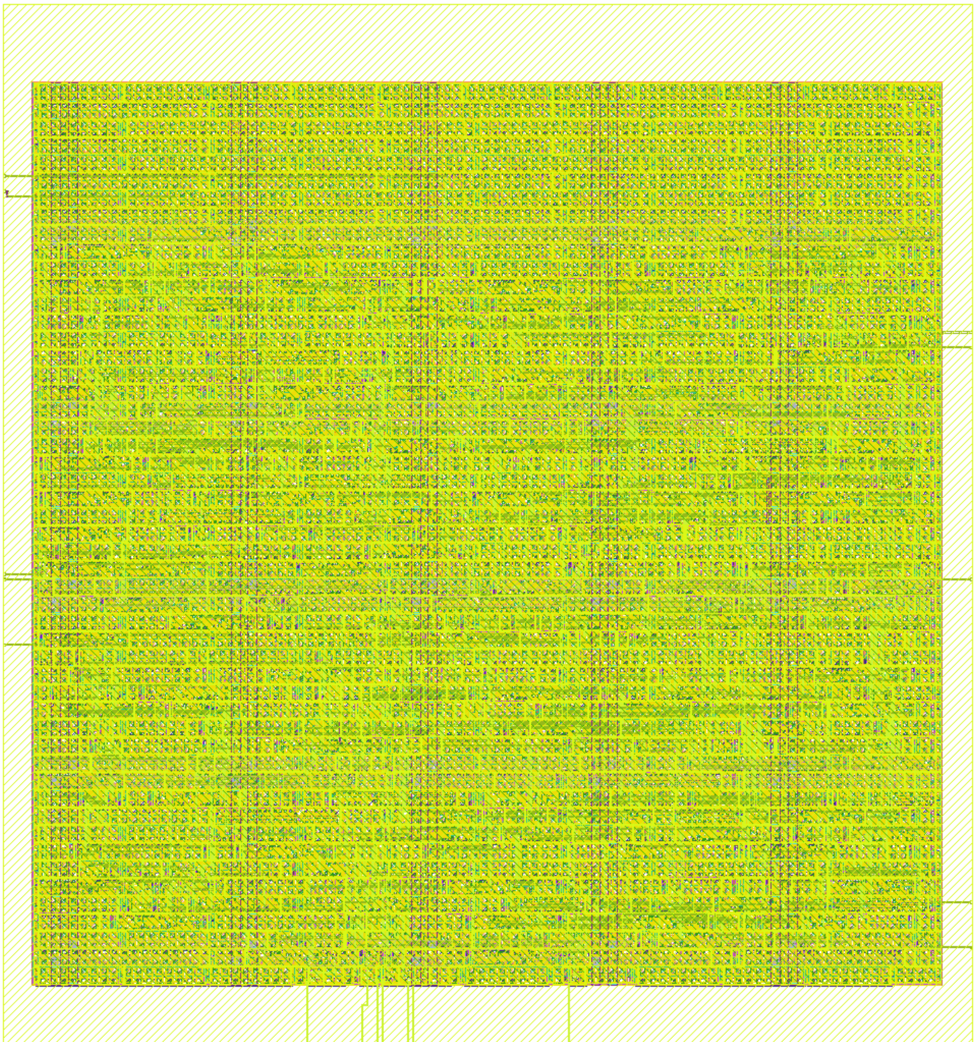
73. code 16, 20, 27, 35 load
74. serial clock data
75. 24 edge decode 4 code
76. frame bit slip

> **PCB** I/O voltage drive strength level shifting ground reference connector pinout pad library PCB prototype RTL

Lab 9 RTL-to-GDS

floorplan placement CTS routing signoff

```
make digital-physical
```



5 *GF180 standard-cell flow* *top*

Metric	
Core/die size	215.49 × 233.41 µm
Die area	50,297.5 µm²
Utilization	39.5%
Standard cells	798 sequential 101
Total wire / vias	14,792 µm / 2,784
Estimated power	6.13 mW

Metric	
digital/asic_digital_top/physical/final_views/	GDS DEF LEF gate-level netlist metrics

10. Signoff

Check		
DRC	0 violations	deck layout rule
Antenna	0 violations	antenna rule
LVS	0 violations	layout netlist
Setup	0 violations	corner setup requirement
Hold	0 violations	corner hold requirement
Worst setup slack	39.21 ns	clock constraint
Worst hold slack	0.336 ns	hold

0 violations tape-out ready library/view constraint

- metrics.json/csv:
- asic_digital_top.gds: foundry
- asic_digital_top.nl.v: gate-level netlist
- resolved.json: flow
- run log/report: warning waiver

- clock period setup slack
- hold slack corner OCV I/O constraint

11.

debug loop

77. commit command
78. testbench cell
79. input internal state output timebase
80. 1 1
81. 1
82. test test
83. commit decision log

SPICE convergence	initial condition ramp discontinuity timestep floating node
code 0/63	comparator polarity reset range
cell	MUX select payload packing
±1	capture convention CDC boundary
RTL sim PASS P&R	clock/reset unsupported construct constraints
DRC/LVS	PDK deck pin label bulk/substrate netlist source

12.

- make check make four-cell-wilkinson
- commit hash command 2
- nominal

B DC

- ramp slope conversion time code scale
- simulation
- VERIFICATION_MATRIX test 1

C CDC boundary

- phase sweep
- binary capture Gray capture failure mode
- ±1 code

D physical experiment

- clock constraint utilization 1
- area slack wire length runtime
- metric metric

README diff command artifact path

13. Prototype tape-out

complete design flow demonstration foundry full chip GDS provider

MPW/provider	run PDK release submission checklist
Analog layout	sampling/MUX/ramp/comparator layout PEX DRC/LVS
PVT/Monte Carlo	corner mismatch pass/fail
Top integration	analog + digital floorplan power/ground clock substrate strategy
Pad/ESD	provider-qualified I/O library voltage ESD latch-up
Package/PCB	bond map package parasitic evaluation PCB FPGA interface
Full-chip signoff	top DRC/LVS antenna ERC density timing deliverable audit

flow GF180 gf180mcu_fd_sc_mcu7t5v0 timing views pad library MPW provider qualified option freeze

Tape-out readiness review

- □ PDK tool version
- □ test artifact
- □ waiver owner
- □ pad ring seal ring density/fill top GDS
- □ package/PCB/test plan chip pinout

14. Specification snapshot

	Prototype v0
Process	GF180MCU Open PDK baseline
Channels	1 analog channel
Storage depth	4 cells
ADC	shared Wilkinson, 6 bit
Readout	24-bit serial payload
Control	synthesizable RTL FSM
Primary objective	complete reproducible design flow
Not yet claimed	target bandwidth, precision, radiation tolerance, production yield

PDK	process device model layer rule deck library
MPW	design wafer cost shuttle
SCA	switched-capacitor array capacitor
Wilkinson ADC	ramp counter ADC
CDC	clock domain crossing
DRC	layout design rule
LVS	layout schematic/netlist
STA	path timing
PEX	layout R/C simulation
GDSII	mask layout

15. Command quick reference

Command	
make check	environment/repository sanity
make nmos-dc	GF180 NMOS DC sweep
make sampling-cell	single sampling cell
make four-cell	4-cell sampling array
make comparator	comparator transient
make comparator-range	input range
make comparator-offset	offset sensitivity
make ramp-generator	ramp
make wilkinson-slice	single 6-bit conversion
make transfer	ADC transfer
make four-cell-mux	shared analog MUX
make four-cell-wilkinson	4-cell integrated analog
make gray-counter	Gray counter RTL
make controller	conversion controller
make four-cell-cosim	SPICE-to-RTL co-sim
make phase-sweep	CDC boundary sweep
make digital-top	full RTL regression
make digital-physical	RTL-to-GDS
make vnc	browser desktop

16. Final review sheet

README prototype complete flow

- ☐ 4-cell/6-bit
- ☐ sampling error ADC code
- ☐ ramp/comparator/counter
- ☐ Gray capture
- ☐ SPICE RTL contract
- ☐ DRC/LVS/STA 0 violations
- ☐ source generated artifact
- ☐ tape-out top-level blocker

analog layout + extracted simulation sampling cell layout schematic ⟶ layout ⟶ DRC ⟶ LVS ⟶ PEX ⟶ transient 1

- GitHub: https://github.com/sykeisuke/asic_rd
- docs/PROTOTYPE_SPECIFICATION.md
- docs/VERIFICATION_MATRIX.md
- docs/TAPEOUT_BLOCKERS.md
- docs/decisions/0003-gray-comparator-capture.md

command commit